ICOTS WORKSHOP HANDOUT ·

Data Dictionary

Student Performance Dataset

A fully explained companion to `student-mat.csv`, with real distributions and a critical-literacy reading of every variable.

Source: Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. UCI Machine Learning Repository. 395 students, two Portuguese secondary schools.

This dictionary exists because a dataset is never just a spreadsheet — it is a set of decisions someone already made about what counts as evidence. Before any model is fit, every column deserves the same question: *what does this actually measure, and what does it stand in for?* Use this document as the handout for Stage 2 (Data) of the ICOTS workshop, or as a standalone teaching resource in its own right.

How to Read This Dictionary

Each variable below is tagged with one critical-literacy category, introduced in Stage 1 of the workshop:

Tag	Category	Definition
HF	Hard fact	Directly observable, low ambiguity — a fact about the world, not a judgment.
SR	Self-report	A perception the student or parent rated, typically on a 1–5 scale — an opinion, not a measurement.
PX	Proxy	Stands in for something not directly measured — often a stand-in for resources, environment, or structural circumstance.
SN	Sensitive	Personal information that deserves scrutiny in a dataset used casually across thousands of ML tutorials worldwide.

A variable can earn more than one tag in spirit. `address` is a hard fact (a student really does live in a rural or urban area) *and* a proxy (for travel time, internet access, and after-school resources). The tag below marks the reading that matters most for critical interpretation, not the only correct answer — disagreement about borderline cases is the point of the Stage 2 sorting exercise.

1 Demographic & Household Attributes

REAL DATA 11 variables describing who the student is and the household they come from. Several of these — especially `Medu`, `Fedu`, `Mjob`, `Fjob` — are the closest thing in this dataset to a measure of socio-economic background, even though none of them says so directly.

Variable	Real distribution	Explanation
`school` HF	GP: 349 MS: 46	Which of the two partner schools (Gabriel Pereira or Mousinho da Silveira) the student attends. A hard fact, but note the huge imbalance — 88% of the sample comes from one school, so any pattern found here may really be a Gabriel-Pereira pattern wearing a general-sounding label.
`sex` HF	F: 208 M: 187	The student's recorded sex. A hard fact on its face, but every model built on this column risks reproducing whatever social patterns already exist in how boys and girls are treated, supported, or disciplined at school — which is exactly why Stage 7 of the workshop audits model fairness across this variable.
`age` HF	Min 15, median 17, mean 16.7, max 22	Age in years. A small number of much older students (up to 22) almost certainly represent grade repetition — meaning `age` quietly encodes some of the same history as `failures` below, just less directly.
`address` PX	R: 88 U: 307	Home address type, rural or urban. This is the dataset's clearest proxy variable: it does not measure travel time, internet quality, or after-school study environment directly, but plausibly stands in for all three at once. Stage 7's fairness audit uses exactly this variable.
`famsize` HF	GT3: 281 LE3: 114	Family size, greater than 3 or less than/equal to 3 members. A simple household fact, though it can proxy for the amount of individual parental attention or household resources available to any one child.
`Pstatus` HF	A: 41 T: 354	Parents' cohabitation status, apart or living together. A hard fact about household structure that is nonetheless very unevenly distributed — only 41 of 395 students (10.4%) have parents living apart, meaning any pattern found for this group rests on a small subgroup, worth remembering during the Capstone Assignment if this is your chosen attribute.
`Medu` SR PX	0–4 scale median 3, mean 2.75	Mother's education level (0 = none, 4 = higher education). Coded as a number, but it is really an ordered category, and it is one of the strongest available proxies in this dataset for household socio-economic resources.

Variable	Real distribution	Explanation
`Fedu` SR PX	0–4 scale median 2, mean 2.52	Father's education level, same scale as `Medu`. Notice fathers' median education is one level lower than mothers' in this sample — worth naming explicitly rather than letting it pass unremarked.
`Mjob` PX	at_home 59, health 34, other 141, services 103, teacher 58	Mother's occupation category. A rough, five-bucket proxy for household income and available time; `other` is the most common answer (36% of mothers), which is itself a sign of how much this variable fails to distinguish between very different real jobs.
`Fjob` PX	at_home 20, health 18, other 217, services 111, teacher 29	Father's occupation category, same scale as `Mjob`. Over half of fathers (55%) fall into the catch-all `other` bucket — more than double the rate for mothers — meaning this column is an even blunter proxy for fathers than for mothers.
`guardian` HF	father 90, mother 273, other 32	The adult who acts as the student's primary guardian. Mothers are guardians for 69% of students in this sample; the 32 students with an "other" guardian are a small enough group that any finding specific to them deserves extra caution.

Table 1: Demographic and household attributes, with real distributions from `student-mat.csv`.

2 School Life & Support

REAL DATA 11 variables describing the student's school-related behaviour and the support structures around them. `failures` deserves special attention here — it is, by a wide margin, the single most important variable in every model fit during this workshop.

Variable	Real distribution	Explanation
`reason` HF PX	course 145, home 109, other 36, reputation 105	Why the student chose this school. On its face a hard fact (a stated reason), but it doubles as a rough proxy for family priorities and expectations — "reputation" and "course" answers likely come from more academically-oriented households than "home" (convenience) or "other."
`traveltime` PX	1–4 scale (1=<15min, 4=>1hr) median 1, mean 1.45	Home-to-school travel time, in four ordered bands. A near-direct proxy for the `address` variable above — most students report short commutes, but the tail of long commutes is worth checking against rural residence.
`studytime` SR	1–4 scale (1=<2hrs, 4=>10hrs) median 2, mean 2.04	Self-reported weekly study time, in four ordered bands. This is the variable most workshop participants guess will dominate the model during the Stage 1 icebreaker — it does not; `failures` and social variables outrank it consistently.
`failures` HF	0–3, median 0, mean 0.33	Number of past class failures. The single strongest predictor in this entire dataset, in every model fit throughout the workshop. It is also, by construction, a record of the very outcome the model is trying to predict, one year earlier — a variable that risks explaining the past more than illuminating the future.
`schoolsup` HF	no 344, yes 51	Whether the student receives extra educational support from the school. Only 13% of students receive this support — worth asking whether that reflects need, availability, or something else about who gets identified for help.
`famsup` SR	no 153, yes 242	Whether the student receives educational support from family. A self-reported yes/no that may reflect the family's ability to help as much as their willingness to.
`paid` HF	no 214, yes 181	Whether the student takes extra paid classes in the course subject (math, for this dataset). A near-direct proxy for household ability to pay for supplementary tuition — 46% of this sample does, a meaningful fraction.

Variable	Real distribution	Explanation
`activities` SR	no 194, yes 201	Whether the student participates in extra-curricular activities. Close to an even split; on its own this variable says little about *which* activities, or how much time they take from studying versus how much they build other skills.
`nursery` HF	no 81, yes 314	Whether the student attended nursery school. A long-ago hard fact that 80% of this sample shares — with so little variation, this variable is unlikely to distinguish much in any model, however meaningful nursery attendance may be in principle.
`higher` SR	no 20, yes 375	Whether the student wants to pursue higher education. A striking 95% answer yes — so little variation exists here that this variable contributes almost no information to any model built on this data, despite sounding like it should matter enormously.
`internet` PX	no 66, yes 329	Whether the student has internet access at home. A fairly direct proxy for household resources and, plausibly, for the same underlying circumstances that `address` and `traveltime` partially capture.

Table 2: School life and support attributes, with real distributions from `student-mat.csv`.

Why does `higher` barely matter, despite sounding important? A variable can only help a model *distinguish* between students if it varies across them. With 375 of 395 students (95%) answering "yes," `higher` is nearly constant — a lesson worth teaching explicitly: a variable's apparent real-world importance and its statistical usefulness in a specific dataset are not the same thing.

3 Social, Lifestyle & Wellbeing

REAL DATA 8 variables describing the student's social life, habits, and self-rated wellbeing. This is where the dataset's most ethically loaded columns live — `Dalc` and `Walc` record alcohol consumption for students as young as 15, in a dataset used casually across thousands of public ML tutorials.

Variable	Real distribution	Explanation
`romantic` SN	no 263, yes 132	Whether the student is in a romantic relationship. Personal information about a minor, self-reported, with no obvious pedagogical justification for its collection — worth asking directly in Stage 1 why a school-performance dataset includes this at all.
`famrel` SR	1–5 scale median 4, mean 3.94	Self-rated quality of family relationships. A textbook example of a number that is not a measurement: two students who both answer "4" may mean substantially different things by it, and the scale has no objective anchor.
`freetime` SR	1–5 scale median 3, mean 3.24	Self-rated amount of free time after school. A perception, not a stopwatch reading — and one that likely correlates with `studytime` and `goout` in ways worth checking before treating all three as independent sources of evidence.
`goout` SR	1–5 scale median 3, mean 3.11	Self-rated frequency of going out with friends. This unassuming variable turns out to be one of the most influential in every model fit during the workshop — it appears in the CART tree's second-level split and ranks highly in the random forest's global importance.
`Dalc` SR SN	1–5 scale median 1, mean 1.48	Self-reported *workday* alcohol consumption, on a 1–5 scale. Recorded for students as young as 15. Most students report the lowest level (median 1), but the mere presence of this column in a dataset this widely used deserves explicit discussion, independent of what the numbers show.
`Walc` SR SN	1–5 scale median 2, mean 2.29	Self-reported *weekend* alcohol consumption, same scale as `Dalc`. Notably higher on average than workday consumption, which is itself an unsurprising but real pattern in the data — worth noting that this is exactly the kind of finding that is easy to report neutrally and easy to forget carries real stakes for real teenagers.
`health` SR	1–5 scale median 4, mean 3.55	Self-rated current health status. Another perception-based scale, useful only with the same caveat as `famrel`: comparability across students is not guaranteed.

Variable	Real distribution	Explanation
`absences` HF	0–75, median 4, mean 5.71	Number of school absences. A hard fact with a long tail — the maximum of 75 absences is far beyond the median of 4, meaning a small number of students with extreme values could disproportionately influence any model that treats this variable as ordinary and symmetric.

Table 3: Social, lifestyle, and wellbeing attributes, with real distributions from `student-mat.csv`.

On `Dalc` and `Walc`. These two columns are not included in this dictionary to normalise their use. They are included, with their real distributions shown plainly, so that the Stage 1 discussion in the workshop — *why does a dataset like this include this at all?* — is grounded in the actual data rather than a hypothetical. Facilitators should feel free to discuss this openly with participants; the goal is critical awareness, not squeamishness.

4 Academic Outcomes

REAL DATA 3 variables recording the student's grades across the school year, each on a 0–20 scale. Every model in this workshop predicts `G3`, or a transformation of it (`pass`, `tier`) — and deliberately excludes `G1` and `G2` from the predictors, for the reason explained below.

Variable	Real distribution	Explanation
`G1` **HF**	0–20, median 11, mean 10.91	First-period grade. A hard, directly measured academic outcome — and a near-perfect predictor of `G3`, since grades tend to be stable across a school year for any individual student.
`G2` **HF**	0–20, median 11, mean 10.71	Second-period grade, same scale as `G1`. Even more predictive of `G3` than `G1`, for the same reason, only closer in time.
`G3` **HF**	0–20, median 11, mean 10.42	Final grade — the target variable for this entire workshop. Transformed into `pass` (`G3` $\geq$ 10) for the main pipeline, and into `tier` (Low/Mid/High) for the Cold Case Files bonus exercise.

Table 4: Academic outcome variables, with real distributions from `student-mat.csv`.

Why drop `G1` and `G2` from every model in this workshop? Because they make the prediction problem close to trivial — a student's grade last term is an excellent predictor of their grade this term, for reasons that have little to do with the more interesting behavioural and social variables this workshop wants participants to reason about. Dropping them is a deliberate pedagogical choice, not an oversight, and it is worth naming explicitly whenever this dataset is reused: a dataset that included `G1`/`G2` in a genuinely deployed early-warning system would need to justify using them directly, since they encode much of what a school likely already knows about a struggling student.

Summary: What This Dictionary Should Leave You With

Thirty-three columns, one spreadsheet, and not one of them is a neutral fact waiting to be discovered. `failures` is a hard fact that also encodes history. `Medu` and `Fedu` are numbers that are really opinions about ordered categories. `address` is a location that is really a proxy for resources. `Dalc` and `Walc` are self-reports about minors that deserve a conversation before they deserve a model. This is the habit of mind Stage 1 and Stage 2 of the workshop are built to teach: read every column as a decision before treating it as a fact.

Category	Tag	Count (of 33)	Share
Hard fact	HF	13	39%
Self-report	SR	13	39%
Proxy	PX	8	24%
Sensitive	SN	3	9%

Counts overlap where a variable carries more than one tag (e.g. `Medu` is both Self-report and Proxy), so shares sum to more than 100%.

Every column is a decision before it is a fact.